

A DATA SCIENCE LAB REPORT

ON

WEB SCRAPING AND DATASET PREPARATION

COURSE CODE: CSE 4114

COURSE TITLE: Introduction To Data Science Lab



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Data Cleaning and Data Insights from CVE Vulnerability Dataset

1. Introduction

This project aims to collect, clean, and analyze cybersecurity vulnerability data using a real-world CVE dataset. The dataset is collected from the Open Source Vulnerability (OSV) database and contains information about vulnerabilities, including CVE IDs, severity, CVSS scores, affected software, CWE classifications, publication dates, and references. The main purpose of this lab is to demonstrate practical data cleaning and data insight extraction techniques.

2. Data Source

The primary data source is **OSV.dev**, an open-source vulnerability database. CVE records are collected using Python-based web scraping and stored in CSV format for further processing.

Source: <https://osv.dev/>

3. Dataset Description

Attribute	Type	Description
CVE ID	String	Unique vulnerability identifier
Severity	String	Vulnerability severity level
CVSS Score	Float	Numerical vulnerability severity score
Published	DateTime	Vulnerability publication date
Modified	DateTime	Last modification date
Summary	String	Short vulnerability description
Details	Text	Detailed vulnerability information
Ecosystem	String	Affected software ecosystem
Package	String	Affected software package
CWE	String	Common Weakness Enumeration
References	Text	Related security references

4. Data Cleaning

The collected CSV dataset will be examined and cleaned before performing analysis. The cleaning process will include removing duplicate CVE records and completely empty rows, handling missing values, removing unnecessary whitespace, and standardizing text fields.

Date fields such as **Published** and **Modified** will be converted into proper datetime format. CVSS scores will be converted into numeric values so that statistical calculations can be performed correctly. CVE IDs will also be checked to ensure that they follow the standard CVE format.

Missing values will be identified and handled appropriately without unnecessarily removing useful vulnerability records. The cleaned dataset will then be saved as a separate CSV file so that the original collected data remains unchanged.

5. Data Insights

After the data cleaning process, exploratory data analysis will be performed to identify useful patterns and trends in the CVE dataset. The Python script will use **Pandas** to calculate statistics and group the data into meaningful categories.

The following insights will be extracted:

- **Dataset Overview:** The total number of CVE records, number of attributes, and overall dataset structure will be identified.
- **Severity Distribution:** The number of vulnerabilities classified as Critical, High, Medium, and Low will be calculated to understand the overall severity distribution.
- **CVSS Score Analysis:** The minimum, maximum, average, and other basic statistics of CVSS scores will be calculated.
- **CVSS Version Analysis:** The dataset will be analyzed to determine how many vulnerabilities use different CVSS versions, such as CVSS v3 and CVSS v4.
- **Publication Trend:** The publication date will be used to determine how many vulnerabilities were reported in each year.
- **Ecosystem Analysis:** Affected software ecosystems will be grouped to identify the most frequently affected ecosystems.
- **CWE Analysis:** CWE identifiers will be analyzed to determine the most common weakness categories found in the dataset.
- **Package Analysis:** Frequently affected software packages will be identified from the dataset.
- **Top Vulnerabilities:** CVEs with the highest CVSS scores will be identified for further examination.
- **Missing Data Analysis:** Missing values in important attributes will be counted to evaluate the quality and completeness of the cleaned dataset.

Simple visualizations will also be generated using **Matplotlib**. Bar charts will be used for severity, ecosystem, CWE, package, and yearly distributions. These visualizations will make the major patterns in the vulnerability dataset easier to understand and present during the laboratory session.

The Data Insight process can therefore be summarized as:

Cleaned CSV → Statistical Analysis → Grouping and Counting → Visualization → Interpretation of Results

6. Dataset Quality and Limitations

The dataset may contain missing information because not every CVE has complete metadata in OSV. Some CVEs may not have a corresponding OSV vulnerability page. During collection, network failures, unavailable records, and server rate limiting may also cause some records to be skipped.

Therefore, the final dataset represents the successfully collected OSV records rather than every CVE published by the CVE program. The insights obtained from the dataset should be interpreted within the limitations of the collected data.

7. Expected Outcome

The expected outcome is a cleaned and structured CVE dataset suitable for analysis. The project will provide meaningful statistical insights into vulnerability severity, CVSS scores, publication trends, CWE categories, affected software ecosystems, and frequently affected packages.

The generated charts and statistical results will provide a simple visual representation of cybersecurity vulnerability trends and demonstrate how raw web-scraped data can be transformed into useful information through data cleaning and exploratory analysis.

8. Conclusion

This lab will demonstrate a complete basic data science workflow using cybersecurity data:

Data Collection → Data Cleaning → Exploratory Analysis → Data Visualization → Data Insights

The final cleaned CSV file, statistical results, and generated charts will be used to demonstrate the practical application of Python, Pandas, and Matplotlib in analyzing real-world cybersecurity vulnerability data.

9. References

1. <https://osv.dev/>
2. <https://www.cve.org/>
3. <https://pandas.pydata.org/>
4. <https://docs.python.org/3/>